

Comparing Numerals 1 to 10

Answer Key

Kindergarten–Grade 1

Quick Answers

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|--|--|
| 1. the symbol = $<$ | 6. the symbol = $>$ |
| 2. the symbol = $>$ | 7. the number that is 1 less than $6 = 5$, the symbol = $>$ |
| 3. the symbol = $=$ | 8. least to greatest = 4, 7, 10 |
| 4. the number that is 1 more than $7 = 8$, the symbol = $<$ | 9. (a) the correct symbol = $<$, — |
| 5. least to greatest = 2, 5, 9 | 10. (a) the correct order = 3, 6, 8, — |
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What is verified

8 of 10 problems fully machine-checked against SymPy.

— marks an answer only you can judge — problems 9, 10. The worked solution states what a correct response must contain.

Curriculum

Level: Kindergarten–Grade 1

K.CC.C.6 – *problems 5, 8, 10*

K.CC.C.7 – *problems 1, 2, 3, 4, 6, 7, 9*

Difficulty 1–4 of 5 across 14 tagged checks.

1. Write $<$, $>$, or $=$ in the blank. 3 _____ 7

Solution: Count up: 1, 2, 3, ... — 3 is said before 7, so 3 is the smaller number. The wide open end of the symbol faces the bigger number, 7. $3 < 7$

2. Write $<$, $>$, or $=$ in the blank. 9 _____ 4

Solution: Counting up, 4 is said before 9, so 9 is the bigger number. The wide end faces 9, which is on the left, so the symbol is $>$. $9 > 4$

3. Write $<$, $>$, or $=$ in the blank. Read both numerals carefully. 6 _____ 6

Solution: Both numerals are 6. Six counters and six counters match one to one with none left over, so neither side is bigger. $6 = 6$

4. Count on 1 from 7, then compare. The number that is 1 more than 7 is _____. 7
_____ 8

Solution: Counting on 1 from 7 gives the next counting number, 8. 8

Now compare 7 and 8: the next number in a count is always the bigger one, so 7 is less than 8 and the wide end faces 8. $7 < 8$

5. Write these three numerals in order from least to greatest. 5 2 9

Solution: Say the counting numbers and pick them off as they come. 2 is said first, so it is least. Next comes 5. Last comes 9, so it is greatest. 2, 5, 9

6. Write $<$, $>$, or $=$ in the blank. 10 _____ 8

Solution: 8 is said before 10 when counting up, so 10 is the bigger number and the wide end faces 10. $10 > 8$

7. Count back 1 from 6, then compare. The number that is 1 less than 6 is _____. 6
_____ 5

Solution: Counting back 1 from 6 gives the number said just before it, 5. 5

Now compare 6 and 5: the number said earlier in a count is the smaller one, so 6 is greater than 5. $6 > 5$

8. Write these three numerals in order from least to greatest. 10 4 7

Solution: Counting up, 4 arrives first, then 7, then 10. A two-digit numeral like 10 counts past every single-digit numeral here, so it is greatest. 4, 7, 10

9. Ivy wrote this and it is not correct: $4 > 9$. (a) Write the correct symbol: 4 _____ 9.
(b) Explain how you know your symbol is right.

Solution (a): Ivy pointed the wide end at 4, but 4 is said before 9 when counting up, so 4 is the smaller number. The wide end must face 9. $4 < 9$

Solution (b) — instructor-judged. A correct response gives a reason, not just the symbol. Any of these earns full credit, in the child's own words: 4 comes before 9 when you count; a group of 4 counters is fewer than a group of 9; on a number line 4 sits to the left of 9; the open end of the symbol has to face the bigger number, and 9 is bigger. A drawing of 4 dots beside 9 dots with the smaller group named also earns full credit. Restating the symbol with no reason earns no credit.

10. Sam put 6, 3, and 8 in order from least to greatest like this: 8, 6, 3. (a) Write the correct order. (b) Explain what Sam did wrong.

Solution (a): Counting up, 3 comes first, then 6, then 8. 3, 6, 8

Solution (b) — instructor-judged. A correct response names the reversal: Sam's list runs from biggest to smallest, but the task asked for smallest first; or, 8 is the greatest number so it belongs at the end, not the start. "He counted backwards" or "he started with the biggest one" earns full credit. Saying only that the order is wrong, without saying why, earns half credit. Rewriting the order with no reason earns no credit here, because part (a) already asks for that.